

Lab 3 - Routing Protocol

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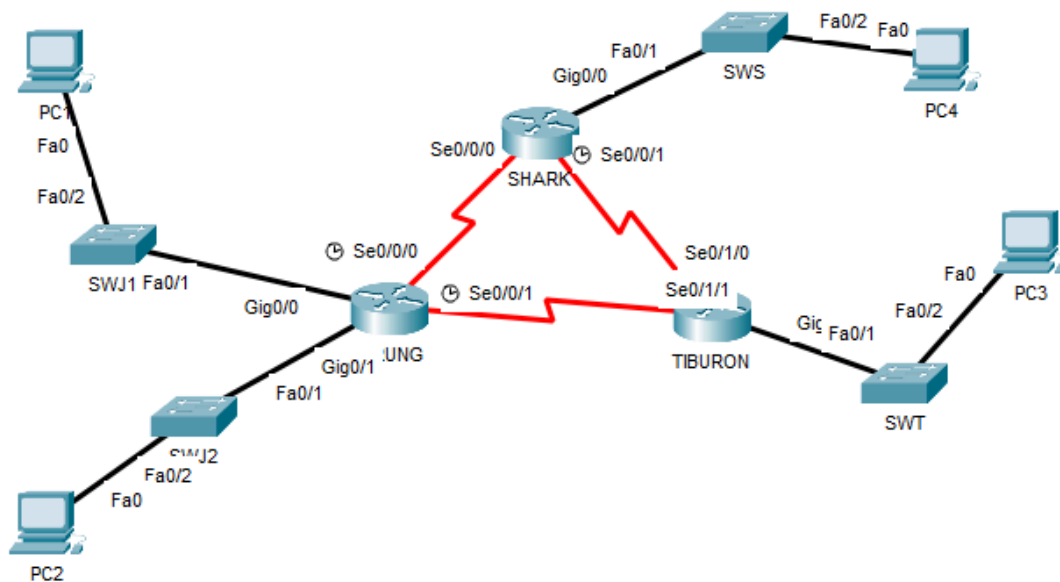
MATRIC NUM.: A23CS0112

SECTION: 02

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.161	255.255.255.224	-
4		G0/1	172.18.110.129	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.65	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.1	255.255.255.192	-
11	PC1	-	172.18.110.190	255.255.255.224	172.18.110.161
12	PC2	-	172.18.110.158	255.255.255.224	172.18.110.129
13	PC3	-	172.18.110.126	255.255.255.192	172.18.110.65
14	PC4	-	172.18.110.62	255.255.255.192	172.18.110.1

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, configure the devices with the appropriate information.

Given IP address
 172.18.110.0/23

172.18.0110 1110 . 0000 0000
 AND 255.255.1111 1111 0.0000 0000
 172.18.0110 111 0.0000 0000

~~172.18.110.0/23~~

SHARK → 60 host

$2^6 = 64$ host

172.18.0110 1110 . 0000 0000
 255.255.1111 1111 1100 0000

172.18.110.0 → Network Address
 172.18.110.1 ~ 172.18.110.62 → Usable Address
 172.18.110.63 → Broadcast Address

255.255.255.192 → Subnet Mask

TIBURON → 50 hosts

$$2^6 = 64$$

172.18.0110	1110.0100	0000
255.255.1111	1111.1100	0000

172.18.110.64 → Network Address

172.18.110.65 - 172.18.110.126 → Usable Address

172.18.110.127 → Broadcast Address

255.255.255.192 → Subnet Mask

JERUNG (LAN 2) → 30 hosts

$$2^5 = 32$$

172.18.0110	1110.0100	0000
255.255.1111	1111.1110	0000

172.18.110.128 → Network Address

172.18.110.129 - 172.18.110.158 → Usable Address

172.18.110.159 → Broadcast Address

255.255.255.224 → Subnet Mask

JERUNG (LAN 1) - 720 hosts

$$2^5 = 32$$

172.18.0110 110. ¹⁰¹~~100~~ 0000

255.255. ¹¹~~111~~ 111 10.111 0 000 0

172.18.110.160 -> Network Address

172.18.110.161 - 172.18.110.190 -> Usable Address

172.18.110.191 -> Broadcast Address

255.255.255.224 -> Subnet Mask

JERUNG - SHARK -> 2 hosts

$$2^2 = 4$$

172.18.0110 110. ¹¹⁰⁰~~1010~~ 00

255.255.1111 111 10.1111 11 00

172.18.110.192 -> Network Address

172.18.110.193 - 172.18.110.194 -> Usable Network Address

172.18.110.195 -> Broadcast Address

255.255.255.252 -> Subnet Mask

JERUNG - TIBURON - 2 hosts

172.18.110.196	0100
255.255.255.1111	1100

172.18.110.196 - Network Address

172.18.110.197 - 172.18.110.198 - Usable Address

172.18.110.199 - Broadcast Address

255.255.255.252 - Subnet Mask

SHARK - TIBURON - 2 hosts

172.18.110.200	1000
255.255.255.1111	1100

172.18.110.200 - Network Address

172.18.110.201 - 172.18.110.202 - Usable Address

172.18.110.203 - Broadcast Address

255.255.255.252 - Subnet Mask

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.

- a. CLI

- i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

JERUNG

```
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.128/27 is directly connected, GigabitEthernet0/1
L       172.18.110.129/32 is directly connected, GigabitEthernet0/1
C       172.18.110.160/27 is directly connected, GigabitEthernet0/0
L       172.18.110.161/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

JERUNG#
```

SHARK

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1
```

SHARK#

TIBURON

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
```

TIBURON#

- b. PT tool
- Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

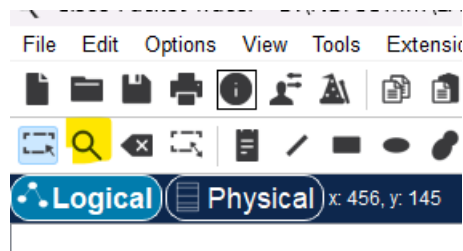


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

SHARK

Routing Table for SHARK				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

TIBURON

Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

```
C:\>ping 172.18.110.158

Pinging 172.18.110.158 with 32 bytes of data:

Reply from 172.18.110.158: bytes=32 time=1ms TTL=127
Reply from 172.18.110.158: bytes=32 time<1ms TTL=127
Reply from 172.18.110.158: bytes=32 time<1ms TTL=127
Reply from 172.18.110.158: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.158:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

- Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

- Explain the reason(s) behind the results.
 - PC1 can ping PC2 because PC1 and PC2 are in the same local area network which is connected with the same router (JERUNG).
 - The result of PC1 pinging PC4 is unreachable because there is no connection between the router JERUNG and other routers (TIBURON and SHARK).
- What needs to be done to ensure all PCs can ping each other successfully?
 - Configure routing for every router.

Task 2 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Routing Table for TIBURON					Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0	C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0	L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```
JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#
```

Figure E

- a. What can you say about the addresses used in the ‘network’ instructions in Figure E?
 - It is for specifying which connected networks the router will advertise the RIP routing.
3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.64/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

- a. What are the changes seen in TIBURON?
 - There is RIP for TIBURON to route to JERUNG networks.
- b. What are the Networks with type R in TIBURON and JERUNG?
 - The networks with type R are the network addresses used to route for other routers. For example, networks with type R in TIBURON are used to route for network addresses in JERUNG.
- c. Ping PC3 from PC1. Was it successful?

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=8ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 8ms, Average = 2ms
```

Yes, it is successful.

- d. Ping PC4 from PC1. Was it successful?

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

No, it is unsuccessful (unreachable).

e. Explain the reasons for your answer.

- Because the RIP for SHARK has not been configured.

f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.0
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
SHARK(config-router)#
```

g. Open router SHARK's routing table and paste here.

Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	
R	172.18.110.64/26	Serial0/0/1	172.18.110.202	120/1	
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1	
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.194/32	Serial0/0/0	---	0/0	
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	
C	172.18.110.200/30	Serial0/0/1	---	0/0	
L	172.18.110.201/32	Serial0/0/1	---	0/0	

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	<pre> C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=2ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms </pre>
2	<pre> C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=8ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=7ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss) Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 8ms, Average = 4ms </pre>
3	<pre> C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Reply from 172.18.110.126: bytes=32 time=8ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 8ms, Average = 2ms </pre>

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0/26
 - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.62
Subnet Mask	255.255.255.192
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?

```
C:\>ping 192.168.1.62

Pinging 192.168.1.62 with 32 bytes of data:

Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.

Ping statistics for 192.168.1.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>ping 192.168.1.62

Pinging 192.168.1.62 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 192.168.1.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

No, PC4 and PC1 ping PC3 is unreachable.

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

SHARK

Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1	
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.194/32	Serial0/0/0	---	0/0	
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	
C	172.18.110.200/30	Serial0/0/1	---	0/0	
L	172.18.110.201/32	Serial0/0/1	---	0/0	

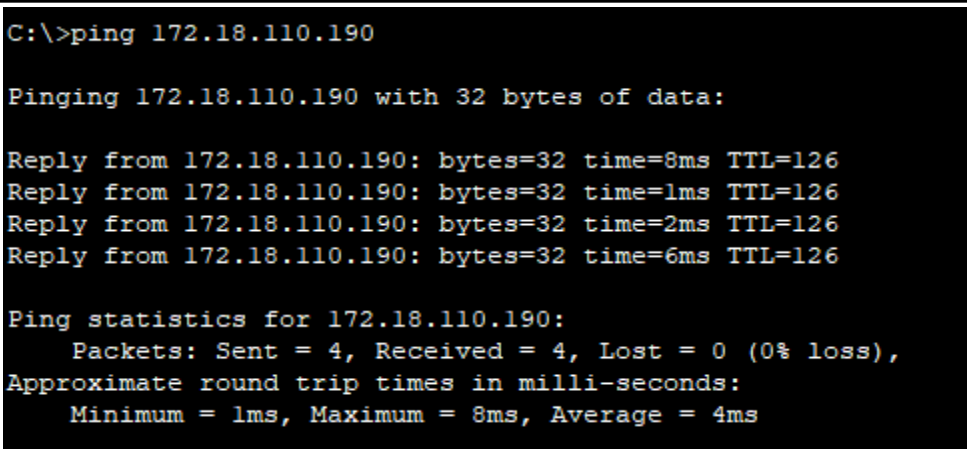
TIBURON

Routing Table for TIBURON					
Type	Network	Port	Next Hop IP	Metric	
R	172.18.110.0/26	Serial0/1/0	172.18.110.201	120/1	
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1	
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1	
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1	
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1	
C	172.18.110.196/30	Serial0/1/1	---	0/0	
L	172.18.110.198/32	Serial0/1/1	---	0/0	
C	172.18.110.200/30	Serial0/1/0	---	0/0	
L	172.18.110.202/32	Serial0/1/0	---	0/0	
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0	
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0	

- e. Referring to the routing table, explain your findings.
- There is no RIP configured for the new network address for PC3.
- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?
- Make sure all the routers have all the RIP configured to all network addresses.
- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON#
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#
```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	 <pre>C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=8ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=2ms TTL=126 Reply from 172.18.110.190: bytes=32 time=6ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 8ms, Average = 4ms</pre>

2	<pre> C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=11ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=9ms TTL=126 Reply from 172.18.110.158: bytes=32 time=5ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 11ms, Average = 6ms </pre>
4	<pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=8ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=10ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 10ms, Average = 5ms </pre>

REFLECTION

What have you learned in this task?

- First of all, I learned how to use more features in Cisco Packet Tracer like configure IP address, check route table and configure routing.
- Second, I learned how to do subnetting for the router and PC.
- Lastly, I learned that routers need to be configured with RIP to ensure end-to-end connectivity.